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SUMMARY

A sequential procedure and a two-stage procedure are proposed for deriving a confidence interval of prescribed width for the unknown truncation parameter of an unknown distribution function. Both these procedures are asymptotically consistent and asymptotically efficient.

1. Introduction

Mukhopadhyay (1974) considered the problem of finding a confidence interval of prescribed width $d > 0$ and coverage probability $1 - \alpha$ for the unknown location parameter θ of an exponential distribution with density function $\sigma^{-1} \exp(-(x-\theta)/\sigma)I(x \geq \theta)$ where σ is also unknown and $I(\cdot)$ denotes the indicator function. Since no fixed sample procedure exists, he proposed a sequential procedure and proved that it is asymptotically "efficient" (i.e. the expected sample size is equal to the smallest sample size possible if σ were known) and asymptotically "consistent" (i.e. covers the unknown θ with probability $1 - \alpha$) as $d \rightarrow 0$.

Let $\{X_n, n \geq 1\}$ be a sequence of independent and identically distributed (i.i.d.) random variables with unknown distribution function $F_\theta(x)$, where $F_\theta(x)$ has a finite but unknown left truncation point θ , that is $F_\theta(\theta) = 0$. Suppose that in some interval $(\theta, \theta + \epsilon)$ $F_\theta(x)$ is twice differentiable with $F''_\theta(x)$ bounded for all x in this interval. Also let $f_\theta(\theta) \equiv \lim_{x \downarrow \theta} F'_\theta(x) > 0$. The problem considered here is to construct

a confidence interval of prescribed width $d > 0$ and coverage probability $1 - \alpha$ for θ .

If $f_\theta(\theta)$ were known this could be achieved (see Chow and Robbins (1965)) as follows:

Define, for $n \geq 1$

$$(1.1) \quad M_n = \min(X_1, X_2, \dots, X_n), I_n = [M_n - d, M_n].$$

Then for a sample size determined by

$$(1.2) \quad n = [(df_\theta(\theta))^{-1} \log \alpha^{-1}] + 1$$

($[\cdot]$ denoting the integer part) the interval I_n has coverage probability

$$P(\theta \in I_n) = P(n(M_n - \theta)f_\theta(\theta) \leq ndf_\theta(\theta)).$$

It follows from Theorem 2.1.5 (p. 56) of Galambos (1978) together with the assumptions on $F_\theta(x)$ that

$$(1.3) \quad n(M_n - \theta)f_\theta(\theta) \xrightarrow{D} Y \text{ as } n \rightarrow \infty$$

where the random variable Y is exponentially distributed with density function $\exp(-y)I(y \geq 0)$. ($\xrightarrow{D}$ denotes convergence in distribution.) From (1.2) it follows that

$\lim_{d \rightarrow 0} ndf_\theta(\theta) = \log \alpha^{-1}$, which together with (1.3) gives

$$\lim_{d \rightarrow 0} P(\theta \in I_n) = 1 - \alpha.$$

For $f_\theta(\theta)$ unknown the fixed sample size in (1.2) cannot be determined and the following sequential procedure is now proposed: Let $X_{n,1} \leq X_{n,2} \leq \dots \leq X_{n,n}$ denote the order statistics of the first n observations. Also, suppose $\{r_n, n \geq 1\}$ is a sequence of integers satisfying

$$(1.4) \quad 1 \leq r_n \leq n, r_n \rightarrow \infty, r_n/n \downarrow 0 \text{ and } r_n/\log n \rightarrow \infty \text{ as } n \rightarrow \infty.$$

Define for each $n = 1, 2, 3, \dots$ the following estimator of $f_\theta(\theta)$:

$$\hat{f}_n = (r_n - 1) \{(n+1)(X_{n,r_n} - X_{n,1})\}^{-1}.$$

In Swanepoel and Van Wyk (1981) a discussion of some properties of this estimator is given. Also, let

$$(1.5) \quad N = \inf \{n \geq n_0 \geq 2 \text{ for which } n \geq (d\hat{f}_n)^{-1} \log \alpha^{-1}\}.$$

The confidence interval for θ is then

$$I_N = [M_N - d, M_N]$$

Theorem 1

The stopping variable N defined by (1.5) has the following properties:

(1.6) N is well-defined and nondecreasing as $d \downarrow 0$,

$$\lim_{d \rightarrow 0} N \neq \infty \text{ a.s. and } \lim_{d \rightarrow 0} E(N) = \infty.$$

(1.7) $\lim_{d \rightarrow 0} N d f_{\theta}(\theta) / \log \alpha^{-1} = 1 \quad \text{a.s.}$

(1.8) $\lim_{d \rightarrow 0} P(\theta \in I_N) = 1 - \alpha$ ("asymptotic consistency")

If there exists a constant $\gamma > 0$ such that

$$(1.9) \quad \sum_{m=m_0}^{\infty} \sup_{n \geq \gamma m} \{ \exp(-c m^2 r_n^2 / n) \} < \infty$$

for every $c > 0$ and some integer m_0 sufficiently large, then

(1.10) $\lim_{d \rightarrow 0} E(N) d f_{\theta}(\theta) / \log \alpha^{-1} = 1$ ("asymptotic efficiency")

Remarks

(i) If $F_{\theta}(x)$ is an exponential distribution function with location and scale parameters θ and σ respectively, Mukhopadhyay (1974) showed that

$$\lim_{d \rightarrow 0} E(N^*) d / \sigma \log \alpha^{-1} = 1$$

where $E(N^*)$ denotes the expected sample size obtained there. This and (1.10) give

$$\lim_{d \rightarrow 0} E(N^*) / E(N) = 1$$

since $f_{\theta}(\theta) = 1/\sigma$ for the exponential distribution.

(ii) Note that for the choice $r_n \sim n^{\nu}$ as $n \rightarrow \infty$, the sum in (1.9) is finite for every $\nu \geq \frac{1}{2}$.

In section 2 Theorem 1 is proved and in section 3 a two-stage procedure is defined with the same asymptotic properties as the sequential procedure given in (1.5).

2 Proof of Theorem 1

The following lemma will be used in the proof of Theorem 1:

Lemma

If (1.9) holds, the set $\{N_d\}_{d>0}$ is uniformly integrable.

Proof

According to a result of Bickel and Yahav (1968) it suffices to prove that

$$(2.1) \quad \sum_{m=m_0}^{\infty} \sup_{0 < d < d_0} P(N_d > m) < \infty$$

for some $d_0 > 0$ and some m_0 sufficiently large. From (1.5) we get for some $\gamma > 0$ large that

$$(2.2) \quad 0 < d < d_0 \quad P(N_d < m) \leq \sup_{n \geq \gamma m} P(X_{n, r_n} - \theta \geq m(r_n - 1)/(n+3) \log \alpha^{-1})$$

Let $U_{n, i} = F_{\theta}(X_{n, i})$. Then $U_{n, 1} \leq U_{n, 2} \leq \dots \leq U_{n, n}$ are the order statistics of i.i.d. random variables uniformly distributed over $[0, 1]$. Using Taylor's theorem we have

$$(2.3) \quad X_{n, r_n} = F_{\theta}^{-1}(U_{n, r_n}) = \theta + U_{n, r_n} / F'_{\theta}(F_{\theta}^{-1}(V_n))$$

for some $0 < V_n < U_{n, r_n}$.

Since we assume $f_{\theta}(\theta) \equiv \lim_{x \downarrow \theta} F'_{\theta}(x) > 0$ it follows that for every constant δ , $0 < \delta < f_{\theta}(\theta)$, there exists a constant $\varepsilon_0 > 0$ such that $F'_{\theta}(x) \geq \delta$ for every $x \in (\theta, \theta + \varepsilon_0)$. From this and (2.3) we therefore obtain

$$\begin{aligned} (2.4) \quad & P(X_{n, r_n} - \theta \geq m(r_n - 1)/(n+3) \log \alpha^{-1}) \\ &= P(X_{n, r_n} - \theta \geq m(r_n - 1)/(n+3) \log \alpha^{-1}, U_{n, r_n} \leq F_{\theta}(\theta + \varepsilon_0)) \\ &\quad + P(X_{n, r_n} - \theta \geq m(r_n - 1)/(n+3) \log \alpha^{-1}, U_{n, r_n} > F_{\theta}(\theta + \varepsilon_0)) \\ &\leq P(U_{n, r_n} \geq \delta m(r_n - 1)/(n+3) \log \alpha^{-1}) + P(U_{n, r_n} > F_{\theta}(\theta + \varepsilon_0)) \end{aligned}$$

Note that $F_{\theta}(\theta + \varepsilon_0) > \delta \varepsilon_0 > 0$. Hence, by application of Theorem 2 of Hoeffding

(1963) we get

$$(2.5) \quad P(U_{n, r_n} > F_{\theta}(\theta + \varepsilon_0)) \leq \exp \{-2n(\delta \varepsilon_0 - r_n/n + 1/n)^2\} \\ \leq \exp(-\delta_0 n)$$

for some $\delta_0 > 0$ and n sufficiently large. In exactly the same way it also follows that

$$(2.6) \quad P(U_{n, r_n} \geq \delta m(r_n - 1)/(n+3) \log \alpha^{-1}) \leq \exp(-cm^2 r_n^2/n)$$

for some constant $c > 0$ and n sufficiently large. The proof of the lemma now follows from (1.9), (2.1), (2.2), (2.4), (2.5) and (2.6).

Proof of Theorem 1

Rewrite (1.5) as

$$N = \inf \{n \geq n_0 \text{ for which } f_{\theta}(\theta)/\hat{f}_n \leq n d f_{\theta}(\theta) / \log \alpha^{-1}\}$$

From (1.4) and Theorem 1 of Swanepoel and Van Wyk (1981) it follows that

$$\lim_{n \rightarrow \infty} f_{\theta}(\theta)/\hat{f}_n = 1 \text{ a.s.}$$

Hence, (1.6) and (1.7) are direct consequences of Lemma 1 of Chow and Robbins (1965). Also the sequence $\{M_n, n \geq 1\}$ is uniformly continuous in probability with respect to the sequence $\{w_n, n \geq 1\}$ where $w_n = 1/n$ (see e.g. Anscombe (1952)). This together with (1.3), (1.7) and Theorem 1 of Anscombe (1952) prove (1.8). Also (1.10) follows directly from (1.7) and the lemma.

3. A Two-stage Procedure

There are situations where a two-stage procedure is more appealing than a sequential procedure. In this section we consider a procedure similar to that of Mukhopadhyay (1980).

Let $n_0 = \max \{2, [(d^{-1} \log \alpha^{-1})^{1/2}] + 1\}$. Start with a sample of size n_0 and compute $\hat{f}_{n_0}$. Then define

$$(3.1) \quad N_0 = \max \{n_0, [(d \hat{f}_{n_0})^{-1} \log \alpha^{-1}] + 1\}$$

and take $I_{N_0} = [M_{N_0} - d, M_{N_0}]$ as interval for θ .

Theorem 2

Under (1.4) it follows that

$$(3.2) \quad \lim_{d \rightarrow 0} N_o d f_{\theta}(\theta) / \log \alpha^{-1} = 1 \text{ a.s.}$$

$$(3.3) \quad \lim_{d \rightarrow 0} P(\theta \in I_{N_o}) = 1 - \alpha \quad (\text{"asymptotic consistency"})$$

$$(3.4) \quad \lim_{d \rightarrow 0} E(N_o) d f_{\theta}(\theta) / \log \alpha^{-1} = 1 \quad (\text{"asymptotic efficiency"})$$

Proof

From (3.1) we have

$$(3.5) \quad (d \hat{f}_{n_o})^{-1} \log \alpha^{-1} \leq N_o \leq (d \hat{f}_{n_o})^{-1} \log \alpha^{-1} + (d^{-1} \log \alpha^{-1})^{1/2} + 4.$$

Therefore, (3.2) and (3.4) follow directly from (3.5). Also, (3.3) follows in the same way as the proof of (1.8).

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